MODEL COMPARISON REPORT

Road Damage Detection | YOLOv8s vs. YOLO11s vs. YOLO26s | 28 October 2026

1. Objective

Compare three fine-tuned YOLO small-scale object detectors and select a candidate for image-based road damage detection deployment. The selection prioritizes validation detection quality while considering model size and inference speed.

2. Experimental setup

All three models were trained in the same Kaggle notebook using the Roboflow road-damage dataset (project ay7agaa/road-damage-l1ju7-jd0gc, version 1). The experiment configured 50 epochs, batch size 16, image size 640, initial learning-rate parameter 0.01, optimizer auto, and seed 42. The recorded runtime used a Tesla T4 GPU.

Validation reported 1,142 usable images with 3,570 labeled instances across seven damage classes: alligator, block, crack, edge, longitudinal, pothole, and transverse. Metrics were computed from the best saved checkpoint for each model.

3. Quantitative results

Model	mAP50	mAP50-95	Precision	Recall	Size (MB)	FPS*
YOLOv8s	0.4101	0.2399	0.6223	0.3979	21.48	59.30
YOLO11s	0.4284	0.2529	0.6122	0.4217	18.30	55.50
YOLO26s	**0.4400**	**0.2578**	**0.6415**	**0.4373**	**19.38**	**57.97**

*FPS values come from 20 repeated predictions on one validation image after one warm-up, as timed in the notebook. Results depend on hardware and benchmark setup.

4. Model selection: YOLO26s

- Highest primary metric: mAP50-95 of 0.2578 (25.78%), ahead of YOLO11s by 0.0049 (0.49 percentage points) and YOLOv8s by 0.0179 (1.79 percentage points).
- Best detection metrics overall: highest mAP50 (0.4400), precision (0.6415), and recall (0.4373) in the comparison.
- Practical balance: its 19.38 MB checkpoint is smaller than YOLOv8s (21.48 MB). Its measured 57.97 FPS is higher than YOLO11s (55.50 FPS).
- Acceptable trade-off: YOLOv8s is marginally faster (59.30 FPS) and YOLO11s is smaller (18.30 MB), but neither matches YOLO26s on the primary accuracy metric.

Decision: Deploy the fine-tuned YOLO26s checkpoint (best.pt) as the candidate demonstration model.

DEPLOYMENT DECISION & CONSIDERATIONS

Road Damage Detection | Model selection notes

5. Deployment implementation

The selected model is deployed as an image inference demo with Ultralytics YOLO and Gradio. The application loads the fine-tuned best.pt file once, accepts an uploaded road image, runs prediction with a configurable confidence threshold (default 0.25) and image size 640, and displays annotated bounding boxes, class labels, confidence scores, and a detection table.

- Input: road-surface image uploaded through the web interface.
- Model: trained YOLO26s detection weights; this is not the unmodified pretrained yolo26s.pt checkpoint.
- Output: marked-up image and a list of predicted road-damage instances.
- Deployment artifacts: app.py, best.pt, requirements.txt, README.md and this report.

6. Interpretation and limitations

- Validation is not a deployment guarantee. mAP50-95 = 25.78% and recall = 43.73% leave substantial room for improvement. Field evaluation is required before use in high-stakes road maintenance or safety decisions.
- Rare class: the notebook reports only one validation instance for the edge class. Its observed performance cannot support reliable conclusions for this class.
- Data quality: notebook logs show some corrupt images and a warning about mixed segmentation and bounding-box annotations. Annotation and dataset consistency should be reviewed.
- Benchmark scope: measured FPS is based on repeated inference of one validation image. End-to-end service latency, CPU performance, diverse images, concurrent requests, and mobile deployment were not benchmarked.
- Selection scope: the reported metrics are validation-set results. An independent test set and representative in-the-wild road images should be used to assess generalization.

7. Next steps

- Evaluate all seven classes on a separate, representative test set and review false positives and missed road damage.
- Expand and rebalance the dataset, especially the edge class, and resolve ambiguous or inconsistent labels.
- Measure CPU/GPU deployment latency, memory usage, and throughput on the actual target hardware.
- Tune the confidence threshold against the required precision-recall trade-off before operational rollout.

8. Conclusion

YOLO26s is the selected deployment candidate because it leads the comparison on the principal metric (mAP50-95), while also achieving the strongest precision, recall, and mAP50, with a reasonable checkpoint size and competitive measured throughput. The Gradio application provides a practical demonstration; further robustness and hardware testing are needed for production use.